

**BEFORE ENGINE START****Familiarise with HANDBOOK first!**

Preflight Inspection ..... COMPLETED

According to Handbook, Cpt. 4.3

Rudder Pedal Position ..... SET

Seat belts ..... TIGHTEN

Flight Controls ..... CHECKED

Canopy ..... LOCKED

Parking Break ..... SET

Apply pressure and set, then 'pump'.

For usage of external Power,  
see notes in Handbook Cpt. 4.4.2**ENGINE START**

Circuit Breakers ..... CHECKED

Backup Battery ..... ON

GTN650Xi, MFD, EFIS ..... BOOTING

**MFD**

Wait for boot sequence.

Check FUEL QTY!

Conf. amber "BACKUP BATT" msg.

Conf. CO value 255 ppm, then decr.

Master Switch ..... ON

**MFD**

Conf. amber "BACKUP BATT" msg.

CAS message OFF

Note down Total Hours from MFD!

Backup Battery ..... OFF

GEN A/B ..... ON

Voltage M ..... MIN. 12 V

(Batt/Alt-C Bus)

Fuel Selector ..... MORE EMPTY TANK

Propeller Control ..... FINE

Fully forward

Throttle ..... CLOSED

Strobe Light ..... ON

LANE A / LANE B ..... ON

Main Fuel Pump ..... ON

– Continues next column!

**CONTINUE: ENGINE START**

Start Power ..... ON (HOLD!)

**LANE A/B Lamps:**

| wait till lamps turn OFF

MFD Engine Indication ..... CHECKED

→ active and correct

→ Fuel Pressure in Limits

**Throttle Lever:**Coolant/Oil Tmp.  $<10^{\circ}\text{C}$   $\geq 55\%$ Coolant/Oil Tmp.  $>10^{\circ}\text{C}$  ca. 45 %

% as indicated as "T" on MFD.

See HINT and Start Performance Chart  
in Handbook Cpt. 4.4.3 on P. 4-8!**ENGINE START:**

Hold "Start Power", THEN press "START"

until Engine fires. Then release

Start-Button and "Start Power".

Throttle ..... 2000 RPM

Oil Pressure ..... CHECKED

LANE A/B Indicator Lamps ..... OFF

Both LANE SWITCHES remain ON!

Only the Indicator Lamps should go out.

Throttle ..... 2500 RPM FOR 10 SEC

To engage/sync generators.

Then back to 2000 RPM

Voltage M / Voltage B ..... 13.4 / 14.0 V

**AFTER ENGINE START**

Avionics ..... ON

ALT-C ..... ON

Voltage M ..... CHECKED

Increasing, nom. 13.8 V

EFIS ..... ON

EFIS Power Source ..... CHECKED

Verify NOT on internal Battery

Baro PFD + MFD ..... SET

StbyEFIS, GTN are synced. MFD isn't!

**BEFORE TAXIING**

Nav Light (at night) ..... AS REQUIRED

Landing-/Taxi-Light ..... ON

A/C moving: Lights on!

PFD, stby EFIS, MFD ..... CHECKED

ADHRS stabilised

**TAXIING**

Keep RPM < 2500, taxiing with walking speed!  
Check BRAKES & STEERING! Ground Control  
CHECKED

Brakes ..... CHECKED

Indications ..... CHECKED

Attitude/direction: PFD, MFD, stby EFIS

**ENGINE RUN UP**

Parking Brake ..... SET

Landing-/Taxi-Light ..... AS REQUIRED

A/C standing still: Lights off!

**Engine Warm-Up:**

| 2500 RPM until oil temp > 50 °C.

Propeller Control ..... FINE

Fully forward

**Pump and Fuel Supply Check**

| Throttle: 3000 RPM

AUX Pump: ON, pressure in limits

AUX Pump Light: ON

Main Fuel Pump: OFF

Pressure still in limits: CHECKED

Main Fuel Pump: ON

Fuel Tank ..... SIDES SWITCHED

Fuel Pressure ..... CHECKED

Fuel Tank ..... FULLEST TANK

– Continues next column!

**CONTINUE: ENGINE RUN UP****LANE Checks**

| Throttle: set to 2500 RPM

LANE A Switch: OFF

Max. vary  $\pm 250$  RPM (aft. 15 sec.)

Check Engine Params in Limits!

LANE A Switch: ON (wait 3 sec.)

LANE B Switch: OFF

Max. vary  $\pm 250$  RPM (aft. 15 sec.)

Check Engine Params in Limits!

LANE B Switch: ON (wait 3 sec.)

**Propeller Check**

| Throttle: set to 4000 RPM

Propeller Control: CHECKED

Cycle 3 times. Move lever SLOWLY  
backwards, but fast forward.

Throttle ..... IDLE CHECK

Throttle ..... SET 3000 RPM

**BEFORE TAKE-OFF**

Instruments ..... CHECKED

Pitot-Heat ..... AS REQUIRED

Flight Controls ..... CHECKED

Cockpit Canopy ..... CLOSED & LOCKED

Seat Belts ..... TIGHTENED

AEPS System ..... UNLOCKED

Store pin in middle compartment!

Engine Instruments ..... CHECKED

**Autopilot**

| Autopilot Switch: ON

Engage, for each test

Check if AP can be overpowered!

Check AP Disconnect by:

Any Trim input, AP-Disc. button.

Autopilot Switch ..... OFF

Autopilot Panel LEDs ..... ALL OFF

Trim ..... T/O POSITION

Flaps ..... AS REQUIRED

Flaps 0 recommended.

Flaps 1 also possible – See AFM!

**TAKE-OFF**

Landing-/Taxi-Light ..... ON

Brakes ..... RELEASED

Propeller Control ..... FULL FORWARD

**When lined-up:**

Throttle ..... FULL POWER

**Engine Instruments:**

| RPM: min. 5700, max. 5800

Manifold Pressure: within LIMITS

Manifold Temp: Below 65 °C

Airspeed Indication ..... COMES ALIVE

**TAKE-OFF SPEEDS**

Nosewheel Unstick ..... 50 KIAS

Lift-Off ..... 60 KIAS

Initial Climb ..... 70 KIAS

Flaps retract ..... 75 KIAS

Not below 150 ft AGL!

Climb .....  $V_Y$  78 KIAS

**CLIMB****Climbing Speeds**

$v_y$  best ROC 78 KIAS

$v_x$  best AOC 63 KIAS

Throttle ..... FULL FORWARD

**Prop Control**

Set Engine Speed

max. 5800 RPM for 5 min

max. continuous 5500 RPM

Instruments ..... T & P IN LIMITS

AUX Pump ..... OFF

Fuel Pressure ..... CHECKED

Landing Light ..... AS REQUIRED

Outside circuit or CTR: OFF

**CRUISE****Fuel Selector**

Switch every 25 min, latest.

Check for balanced quantity!

See notes in AFM Cpt. 4.4.13.

Throttle ..... AS REQUIRED

Prop Control ..... MAX. 5500 RPM

Fuel Flow ..... CHECKED

Pitot Heat ..... AS REQUIRED

**Changes in PWR:**

**Increase Power** ↑ ..... 1. RPM – 2. MAP

**Decrease Power** ↓ ..... 1. MAP – 2. RPM

| PWR   | RPM      | MAP       |
|-------|----------|-----------|
| 100 % | 5500 RPM | 40.5 inHg |
| 75 %  | 5000 RPM | 33.0 inHg |
| 59 %  | 4500 RPM | 25.5 inHg |

**DECENT**

Throttle ..... 15...17 INHG

Prop Control ..... MAX. 5000 RPM

Airspeed ..... 90...100 KIAS

Pitot-Heat ..... AS REQUIRED

Do NOT reduce Throttle to IDLE

in cold OAT environments!

**BEFORE LANDING**

Seat Belt ..... TIGHTENED

Cockpit ..... PREPARED

Autopilot ..... DISENGAGED

Autopilot Switch ..... OFF

Throttle ..... AS REQUIRED

Speed ..... ≤ 83 KIAS

Flaps ..... FLAPS 1

Trim ..... AS REQUIRED

Landing Light ..... ON

AUX Pump ..... ON

**Final Approach**

Flaps ..... FLAPS 2

Speed ..... 65 KIAS

Trim ..... AS REQUIRED

Propeller Control ..... FINE

Fully forward

Throttle ..... SET

Approx. 7 inHG,

Maint. RPM > 2600

**BALKED LANDING / GO-AROUND**

Throttle ..... FULL FORWARD

Speed ..... 70 KIAS

Flaps ..... FLAPS 1

Trim ..... ADJUSTED

Climb .....  $V_Y$  78 KIAS

Follow T/O Procedures

**LANDING**

Throttle ..... CLOSED

Touch-down ..... MAIN WHEELS

Brakes ..... AS NEEDED

**AFTER LANDING**

Flaps ..... RETRACTED

Engine Speed ..... SET

Taxi with walking speed!

Landing-/Taxi-Light ..... ON

Pitot Heat ..... OFF

AUX Fuel Pump ..... OFF

**SHUTDOWN**

Parking Brake ..... SET  
 Landing-/Taxi-Light ..... OFF  
 Propeller Control ..... FINE  
 Engine Speed ..... IDLE  
 Instruments ..... ALL IN LIMITS  
 ALT-C ..... OFF

Check Volts M  $\geq$  13.4 V!

**min. 2 min COOL-DOWN!**

Use timer in PFD or MFD!

Note down Total Hours from MFD!

Lane A ..... OFF  
 after 2 sec, LAMP: ..... CHECKED ON  
 Lane B ..... OFF

**Engine will STOP!**

Main Fuel Pump ..... OFF  
 GEN A/B ..... OFF  
 Avionics ..... OFF  
 EFIS ..... OFF  
 Strobes ..... OFF  
 Master ..... OFF  
 AEPS Locking Pin ..... INSERTED  
 Parking Brake ..... AS REQUIRED  
 All switches are OFF ..... CHECKED

**Parking and tie-down:**

All Lane & Master Switches ..... OFF  
 Parking Brake ..... SET

For short time parking only,  
 for long time, use chocks!

Canopy ..... CLOSED & LOCKED

**AIRCRAFT SECURED!**